Step-by-Step Singular Value Decomposition (SVD) by Hand

Goal

Given a matrix A, we aim to compute its singular value decomposition (SVD):

$$A = U\Sigma V^T$$

Example Matrix

Let

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

This matrix is symmetric and suitable for hand calculation.

Step 1: Compute $A^T A$ and AA^T

$$A^T A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}^T \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}^T = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$$

Step 2: Eigenvalues of $A^T A$

We solve:

$$\det(A^T A - \lambda I) = 0$$

$$\begin{vmatrix} 10-\lambda & 6 \\ 6 & 10-\lambda \end{vmatrix} = (10-\lambda)^2 - 36 = 0$$

$$\lambda^2 - 20\lambda + 64 = 0 \Rightarrow \lambda_1 = 16,\ \lambda_2 = 4$$

$$\sigma_1 = \sqrt{16} = 4, \quad \sigma_2 = \sqrt{4} = 2$$

Step 3: Right Singular Vectors V

For $\lambda = 16$:

$$(A^T A - 16I)v = 0 \Rightarrow \begin{bmatrix} -6 & 6 \\ 6 & -6 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \text{normalized: } \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda = 4$:

$$(A^T A - 4I)v = 0 \Rightarrow \begin{bmatrix} 6 & 6 \\ 6 & 6 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \text{normalized: } \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Step 4: Left Singular Vectors U

We compute:

$$u_i = \frac{1}{\sigma_i} A v_i$$

$$u_1 = \frac{1}{4} A \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u_2 = \frac{1}{2} A \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Step 5: Final Decomposition

$$A = U \Sigma V^T$$

with

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

Verifying:

$$A = U \Sigma V^T = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Conclusion

We successfully decomposed matrix A using the SVD technique by hand. This method allows us to understand the geometry of transformations and is widely used in applications such as data compression, noise filtering, and dimensionality reduction.